Spectral lines and calculation of colour wavelengths and energy level of electrons of hydrogen

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1 Introduction

1.1 Aim

The aim of this experiment is to reliably find the wavelength of light, emitted by the excited electrons of various elements, determining the grating line separation value, as well as calculate the energy emitted by the photons of the excited electrons of the hydrogen atoms.

2 Method

2.1 Set up

Source: https://www.pasco.com/products/complete-experiments/quantum/ex-5546
Apparatus:

- Grating spectrophotometer (such as the one in the image above)
- The following lamps: Sodium lamp, Helium lamp, Neon lamp, Hydrogen lamps
- Device to house and power the various lamps.
- Computer to connect to the spectrophotometer (to record the data collected by the device).

Set up: Spectrophotometer and Lamps
The procedure does not depend on starting with any specific lamp. One may start with any.

1. Connect one of the lamps to the housing unit.
2. Power on the lamp.

3. Let the lamp increase in temperature for approximately a minute or 2 minutes. Note that the lamp can get incredibly hot so handle the lamp with a lot of care. Also note that the lamps can get very bright, so refrain from direct observation upon the lamp.

4. When the lamp is ready, position the lamp near the grating plate. Note that adjustments to the height of the spectrophotometer may be needed depending on the type of lamp or spectrophotometer.

5. Position the rotating arm of the spectrophotometer so that it is initially parallel to the light rays emitted by the lamp[5].

6. On the computer, go into the program and select the "Spectrophotometer", which measures the light intensity with respect to the angle of the rotating arm.

2.2 Method

After the set up is complete, the procedure of the experiment can commence.

1. Slowly move the rotating arm of the spectrophotometer.

2. On the computer, monitor the the light intensity the spectrophotometer is measuring. The rotating arm should start at a large maxima of light intensity. Keep rotating until a 2nd, smaller, maxima is found.

3. Repeat this step a few times to ensure accurate data. Ensure to keep the rotation of the arm as constant as possible.

4. Once satisfactory results of light intensity compared to angle of rotating arm have been collected, save the file onto the computer.

5. Switch the lamp off and wait for the lamp to cool down.

6. Once cool, take the lamp out and replace it with a different lamp.

7. Repeat this process with all the different lamps.

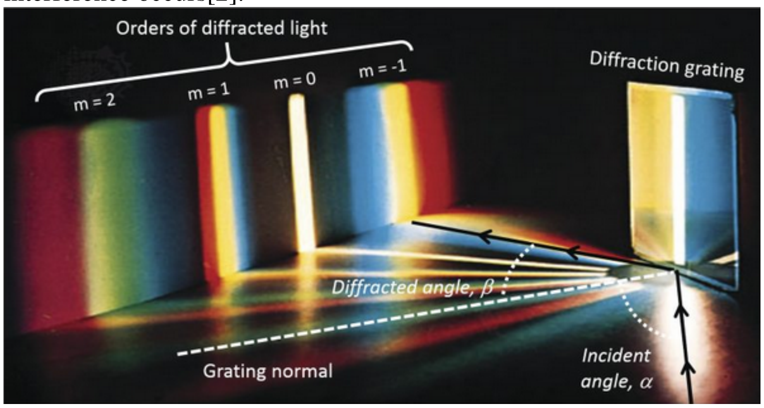
3 Theoretical Background

3.1 Theory

All objects, regardless of their state of matter, emit radiation. This radiation is emitted from the atomic particles that form the constitution of the universe. Typically, excited electrons that have absorbed energy move up from a lower level to a higher level. In order the return back to the ground (lower) state, the electron must then release the energy in the form of a photon with a specific wavelength[1].

Electrons that move up and down between the levels do so at discreet values. For example, in a hydrogen atom if the electron absorbs a photon with a wavelength of 656nm, then it will go up exactly 1 level [6].

Now, because of the particle-wave duality electromagnetic radiation exhibits, constructive and destructive interference can occur when grating lines are introduced to a source of electromagnetic radiation. When the electromagnetic radiation passes through an obstruction, such as a slit in a barrier, a wavefront is created. Typically, multiple wavefronts occur. This then leads to maximas forming, where the constructive interference is at its greatest, and the dark regions or ”fringes” where destructive interference occurs[2].



3.2 Mathematical formulas

When light passes through the diffraction grating, as depicted above, it has a relationship between the angle of the electromagnetic wavefront and the area on the surface the interference is being measured on. The formula that describes this is:

$$dsin(\theta) = m\lambda \tag{1}$$

The value ’m’ describes which maxima is being described. The central maxima is maxima 0. The the first maxima after the central is m = 1, and so on. The value ’d’ describes the separation distance between the lines in the grated surface. The value θ describes the angle between the normal of the surface and the interference being measured.

In the theoretical background, the notion that electrons drop from higher levels to lower levels due that then emit a photnon was mentioned. This phenomena can be described using the following equation:

$$E = -\frac{m_e e^4}{8\epsilon_0^2 h^2}\frac{1}{n^2} \tag{2}$$

With m_e describes the mass of the electron ($9.11 \times 10^{-31}kg$), and e is the electron charge value, and ϵ_0 is the permittivity of free space. Notice that the left fraction is simply a constant value. Hence, it can be simplified into the following equation:

$$E = \frac{-13.6eV}{n^2} \tag{3}$$

Now the photon that is emitted by the electron has an energy value described by the energy lost by the electron, which is:

$$\Delta E = E_f - E_i = (13.6eV)(\frac{1}{n_f^2} - \frac{1}{n_i^2}) \tag{4}$$

Equation 4 will predominantly be used in the hydrogen trial as it is the simplest of the atoms documented in this experiment. To find the wavelength of the emitted photon we use the following equation:

$$\lambda = \frac{hc}{\Delta E} \tag{5}$$

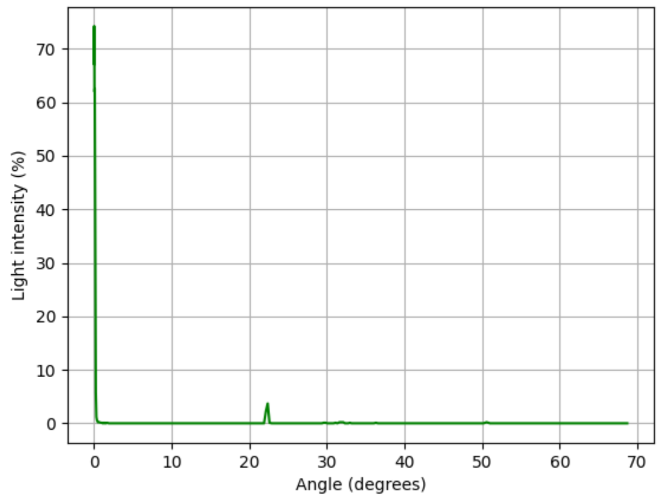
4 Results

The results will follow the following steps:

- The atomic spectra of each element will be displayed as a graph where light intensity is compared to the angle of the rotating arm. This will allow the discovery of first order maxima.
- With sodium, the wavelength of atomic spectra is known, so this trial shall be used to instead calculate grating line separation.
- With Helium, two additional colours within its first order maxima shall be calculated using the calculated grating line separation from the sodium trial.
- With Hydrogen, the same will be repeated as what occurred in the Helium trial, but additionally, the energy of the photons of each wavelength will be calculated.
- With Neon, it will only be displayed.

4.1 Sodium

The following is the results gathered from the Sodium lamp trial



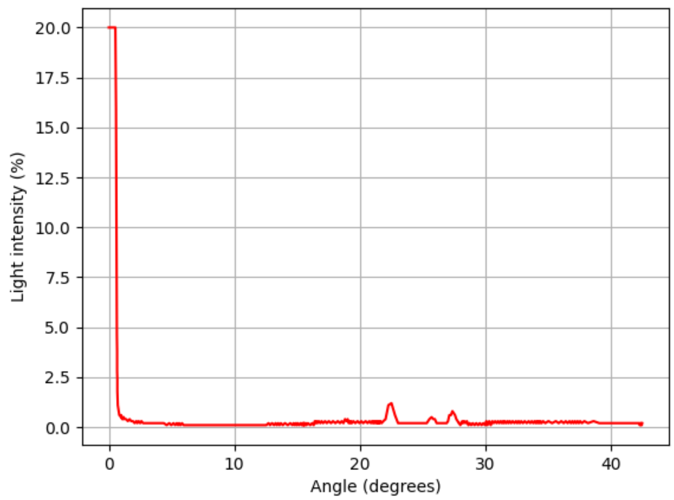
Sodium's light intensity versus angle of rotating arm

The wavelength of sodium's emitted light is given as 589.3 nm. Thus, the grating line separation can be calculated using (1). And the wavelength, as measured on the graph is $\pm 22.35°$

$$d = \frac{\lambda}{sin(\theta)} = \frac{589.3nm}{sin(22.35)} = 1549.71nm \tag{6}$$

This value for the grating line separation shall be used in other calculations throughout the results. The theoretically calculated grating line separation is 1667 nm. This implies that the calculated value deviates with a 7.036% deviation.

4.2 Helium



Graph of Helium's light intensity versus angle of rotating arm
For the calculation of the wavelengths of the colours, we will use the angles $22.045°$ for the one wavelength and $22.515°$, alongside the grating line separation 1549.71 nm. The first colour:

$$dsin(\theta) = \lambda = (1549.71)sin(22.045) = 581.660nm \tag{7}$$

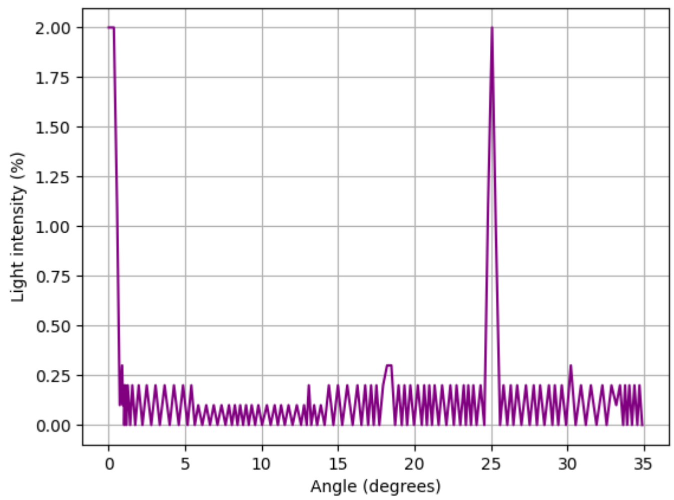
The 2nd colour:

$$dsin(\theta) = \lambda = (1549.71)sin(22.515) = 593.423nm \tag{8}$$

Both of which will fall under the "yellow" spectrum [4]. According to online resourcesn, [3], helium forms a strong maxima at 587.562 nm. With the first calculated wavelength, the deviation percentage sits at a 1.004% deviation. For the 2nd wavelength, the deviation sits at 0.998% deviation.

4.3 Hydrogen

For this example, the same procedure as in the Helium results will be followed, in addition to the energy calculation of the photon.



Graph of Hydrogen's light intensity versus angle of rotating arm

For the calculation of the wavelengths of the colours, we will use the angles $24.600°$ for the one wavelength and $25.475°$, alongside the grating line separation 1549.71 nm. The first colour:

$$dsin(\theta) = \lambda = (1549.71)sin(24.600) = 663.552nm \tag{9}$$

The 2nd colour:

$$dsin(\theta) = \lambda = (1549.71)sin(25.475) = 685.607nm \tag{10}$$

These two wavelengths correspond to the red range of colour [4]. Comparing these values to the values gather online, [7], where the value is 656.2 nm, the standard deviation for the first one is 1.120%. The 2nd wavelength's deviation is 4.481% deviation

Now for the energy calculation, the excited electron must have gone from the 2nd level to the first level due to the structure of the hydrogen atom. Thus the energy calculation, using equations (4) and (5) is:

$$\Delta E = \frac{hc}{\lambda} \tag{11}$$

where, for lambda, we use the angles measured at the first part of this Hydrogen trial of the experiment. That is, $\theta_1 = 24.600$ and $\theta_2 = 25.475$. For the first angle the calculate result is:

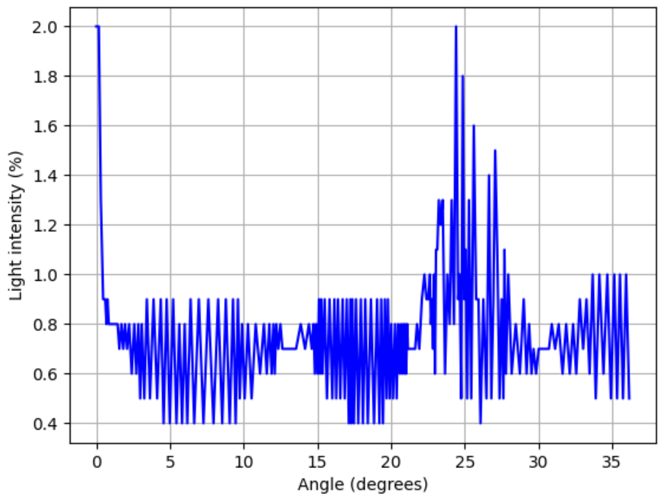
$$\Delta E = \frac{1240eVnm}{(1549.71)sin(24.600°)} = 1.868eV \tag{12}$$

Using the 2nd angle, we get a value of:

$$\Delta E = \frac{1240eVnm}{(1549.71)sin(24.600°)} = 1.808eV \tag{13}$$

4.4 Neon

The following data is just the graph of Neon.



Graph of Neon's light intensity versus angle of rotating arm

5 Discussion

This discussion will focus on each section of the results separately, then fully discuss them together as whole at the end. For sodium, the graph looks very ”plain” in the sense that, the data starts at an incredibly high light intensity value, and then it plummets to 0 (as a result of destructive interference), but then there is a minor ”bump” in intensity as that is taken as the first order maxima. As for the calculated grating line separation value, the deviation sat at a value of 7.036%, which is quite close to the theoretical amount calculated. This reinforces the notion that, at least for this section of the experiment, the data and means of measurement seemed reliable.

For helium, upon looking at the graph, it does closely resemble the graph of sodium, but with slightly more ”noise” in the diagram. After the first order maxima, there are slightly smaller maximas that were detected. This may be due to a faulty measurement taken by the individuals performing the experiment, but overall the graph closely resembles that of sodium. For the wavelengths of the colour calculations, it closely matched that of the researched colour wavelengths. Within an error margin of 0.9 - 1.0 % for each of the wavelengths. This shows quiet high precision for the measurements taken place.

With regards to the hydrogen experiment, the contained significantly more noise, as there were far more oscillations in light intensity detected, and the first order maxima had as much intensity as the centre maxima, which is incredibly irregular. This may

be attributed to incorrect use of the equipment at hand. This erratic results could have lead to larger deviations within the calculations of the wavelengths of colours. While it is still approximately within the range of the wavelength of that colour [4], it did have larger deviations when compared to helium.
With regards to Neon, the results are incredibly erratic as it oscillates far more aggressively than hydrogen. While errors within the handling of equipment could occur by the individuals performing the experiment, this is very drastic and ”chaotic” results for this measurement.

6 Conclusion

This conclusion shall first discuss each component of the aim and how it ties into the experimental results, and then summarise the results and discuss possible implications of the experiment. The aim of this experiment had several components. The first being the calculation of the grating line separation value. The calculation, compared with the theoretical value that was given, resulted in a value that only deviates by approximately 7%, which is not at all an inaccurate answer. Granted more accuracy with experimental could ensure lower deviation values. The same philosophy may be applied to the calculated wavelengths as they too fell quite close to the theoretically calculated values found online.

However, it must be noted that the Neon graph came out incredibly erratic, due to either a technical problem with the measurement devices, or an error in using said equipment. Either way, it exhibits that while the data of the other values can still be fairly accurate, it can go wrong for a reason that may not be clear at first.

Overall, the experiment still shows incredible accuracy when data is recorded correctly, and reinforces the current understanding of atomic models and atomic emission spectra as well as energy emitted by electrons moving between orbital levels.

7 Bibliography

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